

The block gallery

Every function a `design.typ` calls, rendered once, so a change in any one of them is visible in a diff of this document.

How this file is read

A section is the unit: a muted lead, one visual, then the body.

HOW TO READ THIS



a part this document owns



a part owned elsewhere, drawn as a pointer



a relationship inside this unit



a relationship crossing the seam

captured

a coverage row marked captured

standard

a coverage row marked standard

out-of-scope

a coverage row marked out-of-scope

A design document calls these functions directly. The author writes words and structure; every visual decision lives in the library, so nothing here is styled at the call site.

- A bullet holds one property.
- Three sentences is the budget. A fourth is a new bullet.
- Plain prose beside bullets is legal, and better when the sense is a chain of causes.

The foundation statements

Four kinds share one anatomy, so a reader learns the shape once.

GOAL

Author content, never styling

The authored file carries words and structure. A reviewer reading a diff reads the claim, not the presentation.

composition

NO-GOAL

A general document system

This renders design layers. A different kind of document is a different tool's problem.

PRINCIPLE

One name per concept

A block kind and the function rendering it carry the same name, so there is nothing to translate and nothing to look up.

modeling

composition

Enforcement and evidence

INVARIANT

Every diagram edge names a declared node

The library checks each endpoint against the declared node ids, so an edge to nowhere stops the compile instead of rendering a missing line.

mechanism

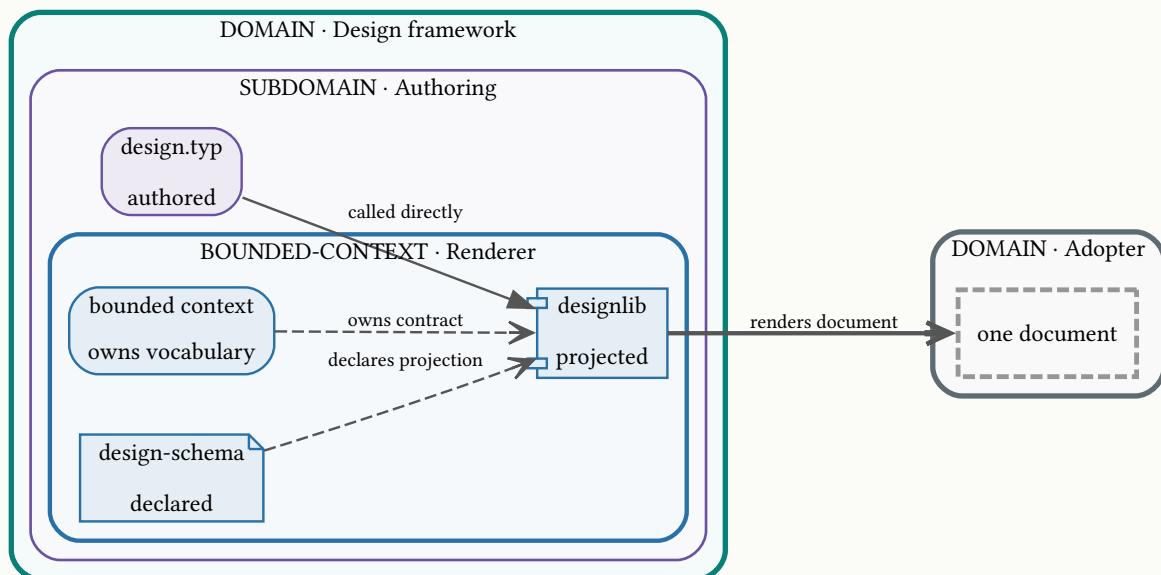
ON ENFORCEMENT LABELS

The enforcement argument is required. A convention labelled honestly beats a mechanism claimed falsely, and an omitted label claims nothing while looking like a claim.

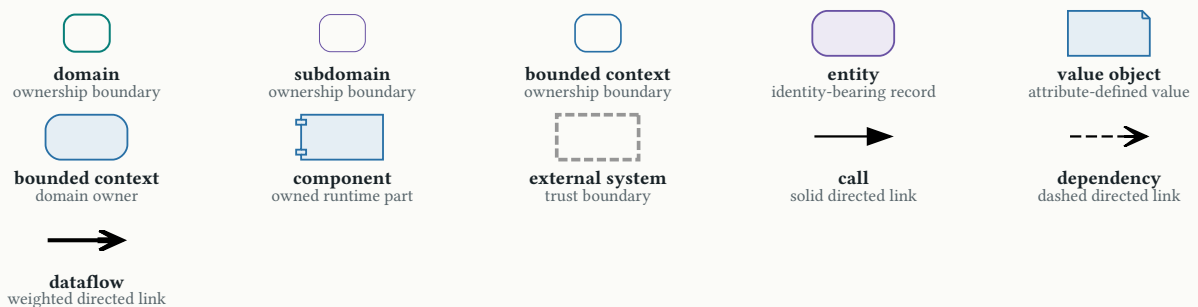
The diagram

Nodes, semantic boundaries, and typed relations are data; the solver lays out this graph.

VIEWPOINT CONTEXT-OWNERSHIP · ALTITUDE L2 · CONTEXTS the authoring surface over one library



LEGEND



A design document calls the library directly, so there is no conversion step between what was written and what is checked. Boundary labels name what each enclosing group means.

Altitude states the depth. Viewpoint states the question answered at that depth. A group declares one semantic boundary, a node kind selects its framework architecture role and Graphviz shape, and a dictionary edge declares one directional relation. Graphviz solves placement but does not define those roles. Separate diagrams do not share an identity registry.

The positioned diagram

Coordinates carry this drawing's reading order.

ALTITUDE L3 · COMPONENTS the explicit left-to-right path



Use the positioned carrier only when an authored grid is part of the meaning or precise visual control is necessary.

Units and their answers

A component answers five questions; absent answers stay absent.

projector

composition

Turns the declared schema into the library.

Primary component prose uses the renderer body size.

responsibility

One schema in, one library out.

failure

Refuses a schema missing a required vocabulary.

library

invariants

Holds every block contract as a signature.

responsibility

Validate one block, then render it.

invariants

An enum value outside its declared set panics.

aggregate

state

Assembles the contexts into one document.

schema

Declares the vocabulary.

renderer

Draws the declared structure.

gate

Checks the rendered document.

The gate

responsibility

Decide pass or fail on a whole layer.

interface

Three commands: project, aggregate, check.

interactions

Reads the schema; runs the renderer.

invariants

A stale rendered document fails the check.

failure

Exit 1 on a violation, exit 2 on a missing tool.

The projection contract

contract composition

Keep the declared schema and projected library aligned.

responsibility

Project the schema into the callable library.

interface

The schema and library compile as one contract.

failure

A projection mismatch stops the render.

Magnitudes and breadth

A tile states a fact; a coverage row states a decision.

2.4M

words rendered

+12%

93

gate assertions

3

silent failures caught

-2

Coverage

Every row carries a status, and a row that is not captured states why, so an absence is a recorded decision rather than an oversight.

PART	STATUS	WHY
scripts/design-aggregate	captured	Rendered by this document.
flake.nix	standard	Conventional Nix packaging.
LICENSE	out-of-scope	Not a designed subsystem.

Pending updates

build The context map is still authored by hand

2026-08-23

The context map and the coverage map are markdown files kept beside the layer they index, so each can disagree with what it indexes. Generating them from the Typst sources would remove the disagreement rather than check for it.

ADR-0065

verify The mechanism fence catches only a denylist

2026-08-23

A behavior clause must state an observable outcome rather than the mechanism producing it. The check matches a list of implementation-shaped words and a few id patterns, so it is labelled partial: a mechanism described in ordinary words passes it.

Cross references

A reference is a call, so a rename is a compile error.

A context is named with `design-layer`, or explicitly tinted with `design-layer`, a term with `term-pending-ledger`, and a judgement axis with `robustness`. A decision is cited by number — `ADR-0065` — and the number is the whole citation: the gate resolves it against the files on disk, so a citation of a decision that does not exist is caught at the citing line, where a hand-typed path would rot silently on a rename.

Each of those is a `chip` — the one inline primitive they share, tinted by what it names. Written bare it carries no tint, which is the honest rendering of a token that names no declared vocabulary.

A judgement axis also renders as a standalone `robustness` pill, and a combination of two sorts itself into the declared enum order: `modeling composition`. An external destination is written as a `link` that carries its own tone.

The entity census

An entity states what it is made of and how it sits against others.

Design layer aggregate stateful documentation owned by the author

The authored model and the document rendered from it.

ATTRIBUTES

authored **Notation** · Authoring notation
The notation the layer is written in.

derived **Rendered document** · Design document
The rendered document, produced by the renderer and never hand-edited.

RELATIONSHIPS

1 : 0..n A layer holds many contexts, and a context belongs to exactly one layer.

Rendered layer entity append-only documentation owned by the renderer

The document produced from the authored layer.

ATTRIBUTES

derived **Pages** · Rendered pages
The pages produced by the renderer.

The two groups label themselves from the run of clauses, so the authored source names neither heading.

Behavior rules

A rule is a context, one event, and the outcomes a reader can observe.

BEHAVIOR

A stale rendered document fails the check

Given The layer's sources have changed since the document was rendered.

When The gate runs its freshness check.

Then The check reports the document as stale and exits non-zero.

Then No rendered artifact is written, so nothing reads as having passed.

level: interface area: Mechanical repository commands

Admonitions, cards, and tables

Each shape carries one job, and the tint says which.

WHAT A GATE PROVES

A freshness check proves render-matches-render. It never proves render-matches-source, and stating the limit is what keeps the claim honest.

A DEGRADING CHECK

A check whose helper is missing can report SKIP and still exit 0. That is worse than no check, because it reports success.

Invariant

A rule whose violation makes the model wrong. It panics. This long explanation keeps the explicit two-column contract visible even when automatic layout would prefer one column.

Guideline

A rule of style. It is silent unless the author asks for it. This long explanation keeps the explicit two-column contract visible even when automatic layout would prefer one column.

Phase	Reads	Decides
assert	the authored source	document-level contracts
compile	the generated typst	per-block contracts
freshness	the committed document	whether it is current

PROSE WITH A COMPACT STATUS

Explanation	Status
A compact final column leaves the remaining measure to the explanation.	ready
A second row proves the divider without closing the outside edge.	held

FIVE-COLUMN DECISION MATRIX

Concern	Owner	State	Consequence	Record
Rendering	library	active	The table fills the measure while compact columns retain content width.	ADR-0123
Validation	gate	required	Malformed rows fail before the renderer can imply missing cells.	ADR-0124

SINGLE-COLUMN REGISTER

Declared behavior

The single track consumes the complete measure.

The first row remains a semantic table header.

CODE

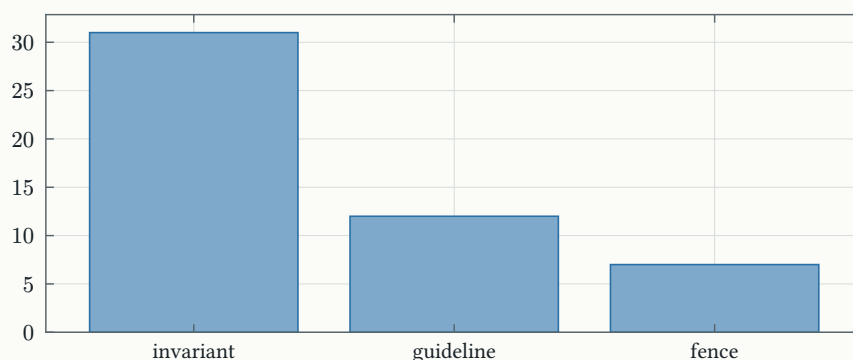
```
nix run .#check -- docs/design .
```

BASH

Plots and figures

A chart states a magnitude; a figure carries renderer markup.

CHART · COLUMN rules by kind



A column chart reads its categories across the bottom.

CHART · BAR coverage rows by status

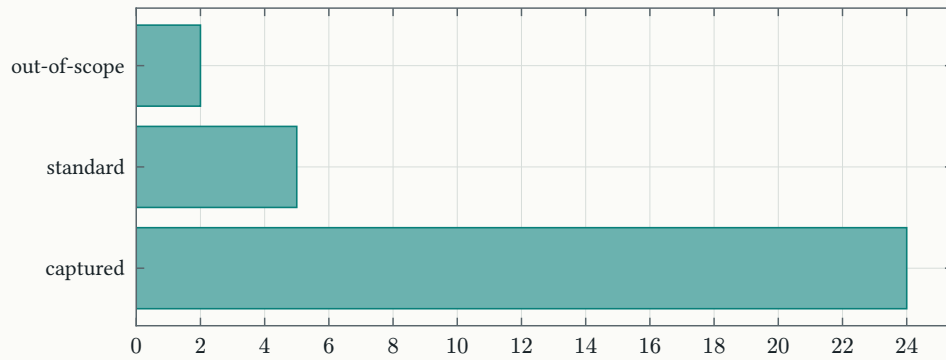


CHART · LINE decisions recorded, by version

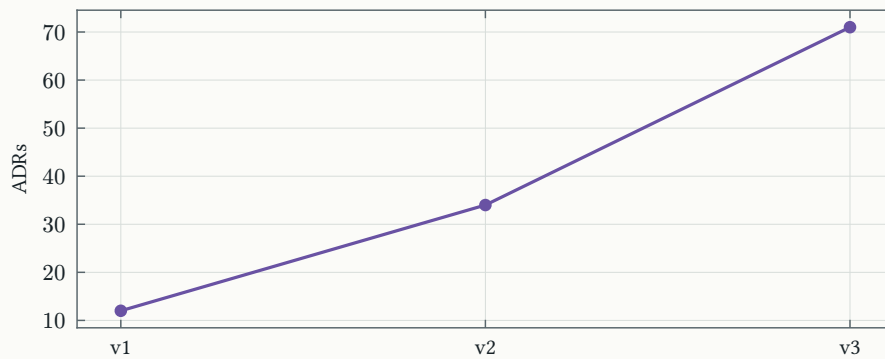
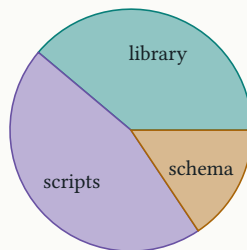


CHART · PIE where the lines live



NATIVE FIGURE · cetz

a drawing would sit here

A figure names the vendored packages it uses, so a render stays offline-reproducible.

EXTERNAL SVG

diagram.svg

An external drawing

Equations and formulas

Native mathematics carries the same hierarchy and reading aids as every other design unit.

Inline mathematics remains part of prose: the availability target is $A \geq 0.999$. An expression that needs a name, notation key, caption, or stable target becomes a formula block.

FORMULA	Availability budget
$A = 1 - \frac{D}{T} \tag{1}$	
NOTATION	
A availability over the observation window	
D accumulated downtime	
T total observation time	
<i>The expression makes the shared reliability budget explicit; subsystem targets may refine it but cannot silently redefine it.</i>	

Implementation-neutral procedures

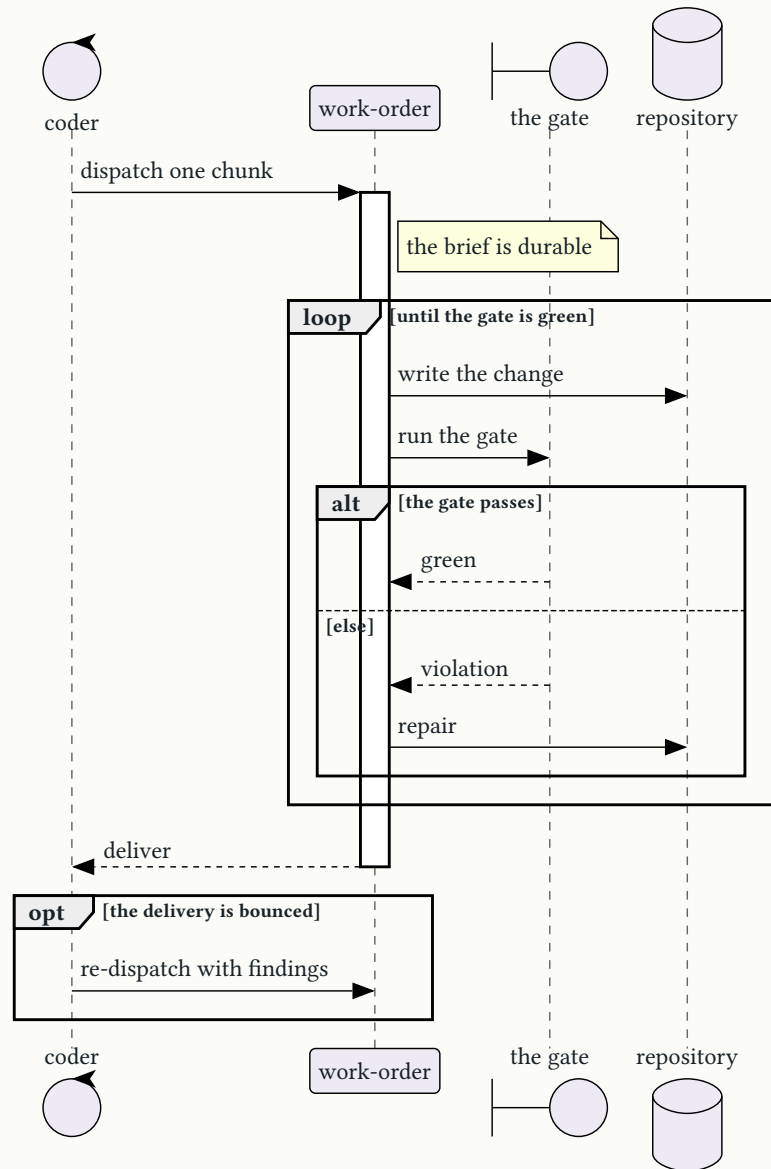
Pseudocode makes consequential control flow explicit without choosing a programming language.

PSEUDOCODE	Accept one coherent change				
<table><tbody><tr><td>INPUT</td><td>a durable work order · the current repository state</td></tr><tr><td>OUTPUT</td><td>an accepted change · actionable findings</td></tr></tbody></table>		INPUT	a durable work order · the current repository state	OUTPUT	an accepted change · actionable findings
INPUT	a durable work order · the current repository state				
OUTPUT	an accepted change · actionable findings				
<pre>DO read the design brief and its acceptance evidence FOR EACH criterion IN the acceptance criteria IF the criterion can be observed DO bind the criterion to one check OTHERWISE RETURN a clarification request END IF END FOR REPEAT DO implement one coherent change DO run the complete gate UNTIL the gate is green RETURN the accepted change</pre>					
<i>Keywords, indentation, branches, and loops belong to the renderer; the author supplies only the procedure's meaning.</i>					

The sequence

A sequence is ordered by time, and says when a party is active.

SEQUENCE a work order, from dispatch to accepted change



Activation, branch, repetition, and an optional block — the constructs a node-and-edge drawing cannot state.

The state machine

A machine declares states and transitions, never positions.

Issue state needs_triage needs_info ready_for_agent in_progress done wontfix

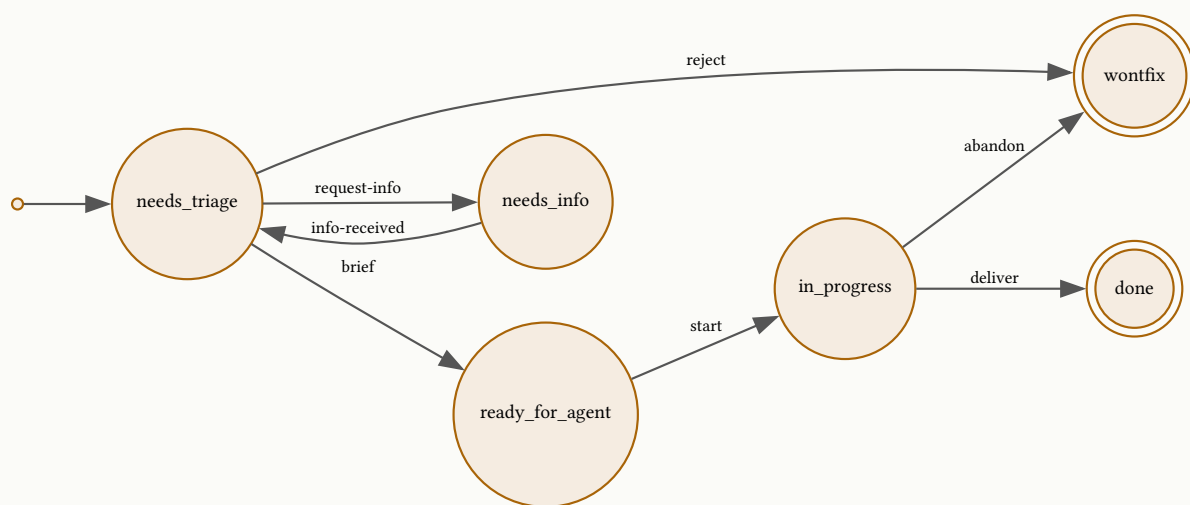
Issue entity stateful delivery owned by the tracker

A tracked unit of work.

ATTRIBUTES

derived State · Issue state `issue-state` · `issue-lifecycle`
The issue's current lifecycle state.

STATE MACHINE an issue, from intake to its two terminals



The carrier solves the layout, draws the initial marker, and rings each accepting state.

End-to-end walkthrough

A prose-only section is legal, and this one demonstrates it.

An author writes a `design.typ` beside its `CONTEXT.typ`. Each file is a module exporting a title and a body, so nothing in it decides page size, numbering, or colour.

The aggregate discovers each context, imports it as a module, and places its body under a chapter heading. The result is one document with one table of contents, and the gate proves it is current by rendering it again and comparing the bytes.

A section that reaches for narrative rather than a diagram is not a failure of the document. The spine mandates a walkthrough, and a walkthrough is a sequence carried in sentences, so the visual argument is optional and this section supplies none.

Generated from the projected library — never hand-styled.